PEER REVIEW HISTORY

BMJ Paediatrics Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence of traumatic psychological stress reactions in children and parents following paediatric surgery. A systematic review and meta-analysis.
AUTHORS	Turgoose, David Paul Kerr, Stephanie De Coppi, Paolo Blackburn, Simon Wilkinson, Simon Rooney, Natasha Martin, Richard Gray, Suzanne Hudson, Lee Duncan

VERSION 1 – REVIEW

REVIEWER	Reviewer name: Dr. Peter Flom Institution and Country: Peter Flom Consulting, United States Competing interests: None
REVIEW RETURNED	02-May-2021

GENERAL COMMENTS	I confine my remarks to statistical aspects of this paper. The general approach is fine, but I have a few issues to resolve before I can recommend publication. p 5 line 15 is this 15% of all kids, or 15% of those who experience a trauma? p. 6 line 47 "validated" in what way? also, were there language requirements? Figures - please move the solid line to the left and the numbers to the right, giving a lot more space to the actual figure. Also, label the dotted line with "mean" (or similar). Peter Flom

REVIEWER	Reviewer name: Dr. Andrew Holland Institution and Country: Competing interests: None
REVIEW RETURNED	14-May-2021

GENERAL COMMENTS	Thank you for asking me to review this manuscript. The authors have performed a systematic review of publications reporting ASD and PTSD in children and their parents following surgery. Unsurprisingly, the authors conclude that there is a high prevalence of both conditions in both groups following surgery. I have the following comments:

	1. The authors have not defined what they mean by 'surgery'. There would likely be a difference between ligation of a skin tag under local anaesthesia versus a Fontan procedure, for example. Would any procedure requiring a GA be considered 'surgery': for example, an upper GI endoscopy? 2. Was there any difference between GA versus LA 'surgery'? 3. Was there any difference between day-case as oppose to surgery that required one or more nights in hospital? 4. Did any of the studies assess pre-surgery levels of anxiety and correlate this with ASD/PTSD? In my experience extremely anxious parents tolerate surgery in their children very poorly (and the child also experiences more pain and more frequent complications) and would be expected to suffer more ASD/PTSD. 5. The statistical analysis was complex. I personally am not familiar with the "Freeman-Tukey double arcsine transformation" for example. I would recommend an independent assessment by a statistical reviewer. 6. There is no control group. The authors have stated ASD/PTSD was nigher than reported rates in the general population, but it would be useful to know if the rates were higher than those of parents and children presenting to a Emergency Department or medical inpatients not requiring surgery. 7. The authors suggest that this study may help raise awareness of this issue. Whilst I agree, perhaps a more practical suggestion would be t recommend a screening questionnaire or other method that could be used to identify those children and/or families at risk of more significant ASD/PTSD so that targeted interventions could be used. Most healthcare systems have significant resource constraints and have to be able to husband these appropriately.

VERSION 1 – AUTHOR RESPONSE

Dear editors,

Thank you for asking us to submit revisions for your consideration. Thank you also to the reviewers for important questions to improve our paper.

We have dealt with each request in turn:

1. Title page : Kindly make sure that the title page should be embedded at the first page of your main document.

This has now been done

2. Figures below 300 dpi: Please ensure that your figures are a minimum of 300 dpi and a maximum of 600 dpi. For online reviewing we do not require print quality images and the larger file sizes can slow down the running of the review process.

Apologies for this. We have now altered each image to have a dpi of 300.

3. Title delete "What is the" and "?"

Thank you we have changed this.

Replace Key messages with What this study adds ans What is already known. SRs are original articles not reviews.

Thank you we have changed this.

Discussion 1st sentence delete "We believe that this is the first ". Journal style is to AVOID use of describing the study as the first (see instructions to authors)

Thank you we have changed this.

Table 1 would be better as two or three separate tables: cardiac studies in one; mixed types of surgery in another; the rest in another.

Thank you we have kept one table, but we have separated out into three sections to reflect the different surgery types as suggested.

Associate Editor
Comments to the Author:
Please provide a completed PRISMA checklist with your revision, indicating where and how you have complied with all the individual component requirements. If there are parts where you have not yet complied please ensure that you do so in your revision.

Thank you. We have complied with all PRISMA criteria – the one exception being registration. We acknowledge that we did not register the systematic review with an organisation such as PROSPERO. We hope that this wouldn't preclude publication. The checklist has been updated accordingly.
Reviewer: 1
Dr. Peter Flom, Peter Flom Consulting
Comments to the Author
I confine my remarks to statistical aspects of this paper. The general approach is fine, but I have a few issues to resolve before I can recommend publication.

p 5 line 15 is this 15% of all kids, or 15% of those who experience a trauma?

Thank you we have amended this to provide greater clarification.

p. 6 line 47 "validated" in what way?

Thanks for asking for clarification. By this we mean questionnaires which have been used in included studies but have been previously been established as formally validated rather than a new questionnaire invented for a study –thus they have been tested as valid measures previously.

also, were there language requirements?

Thank you for this important clarification – we decided to include papers of all languages a priori. As it happens, we didn't find any papers in non-English. Accordingly, we have amended the detail on inclusion criteria

Figures - please move the solid line to the left and the numbers to the right, giving a lot more space to the actual figure. Also, label the dotted line with "mean" (or similar).

Thank you. We have redone the forest plots to fit this recommendation within the constraints of what was possible with the statistics package. We have also added a change in the labelling of the figure to highlight the meaning of the dotted line.

Peter Flom

Reviewer: 2
Dr. Andrew Holland
Comments to the Author
Thank you for asking me to review this manuscript. The authors have performed a systematic review of publications reporting ASD and PTSD in children and their parents following surgery. Unsurprisingly, the authors conclude that there is a high prevalence of both conditions in both groups following surgery. I have the following comments:
1. The authors have not defined what they mean by 'surgery'. There would likely be a difference between ligation of a skin tag under local anaesthesia versus a Fontan procedure, for example. Would any procedure requiring a GA be considered 'surgery': for example, an upper GI endoscopy?

Thank you for this clarification. In our search we sought any type of surgery. Accordingly, we have been clear about this in the methods by highlighting any type of surgery in inclusion criteria. We didn't include any medical procedure – e.g. endoscopy, but the intervention in the study itself had to describe itself as surgery.

2. Was there any difference between GA versus LA 'surgery'?

We didn't find any papers looking at psychological trauma after local.

3. Was there any difference between day-case as oppose to surgery that required one or more nights in hospital?

One paper – Scrimin et al. (2009) – did look at day surgery. We have explored the role of type pf surgery and length of stay within the results and discussion. There was insufficient information across studies to use this information in meta-regression as a predictor.

4. Did any of the studies assess pre-surgery levels of anxiety and correlate this with ASD/PTSD? In my experience extremely anxious parents tolerate surgery in their children very poorly (and the child also experiences more pain and more frequent complications) and would be expected to suffer more ASD/PTSD.

We agree that this is an important factor. We discuss in the discussion as a common limitation in studies identified and included, and we also made a recommendation for further research that baseline and post measures should be made.

5. The statistical analysis was complex. I personally am not familiar with the "Freeman-Tukey double arcsine transformation" for example. I would recommend an independent assessment by a statistical reviewer.

We believe that the other reviewer has provided a statistical review.

6. There is no control group. The authors have stated ASD/PTSD was nigher than reported rates in the general population, but it would be useful to know if the rates were higher than those of parents and children presenting to a Emergency Department or medical inpatients not requiring surgery.

We agree that there was a lack of case control studies. Indeed, our meta-analysis was restricted to prevalence rather than comparisons between cases and controls. This is referred to in the discussion.

7. The authors suggest that this study may help raise awareness of this issue. Whilst I agree, perhaps a more practical suggestion would be t recommend a screening questionnaire or other method that could be used to identify those children and/or families at risk of more significant ASD/PTSD so that targeted interventions could be used. Most healthcare systems have significant resource constraints and have to be able to husband these appropriately.

Thank you. We have amended the conclusion section to emphasise this.

Your sincerely,

David Turgoose

Lee Hudson

VERSION 2 – REVIEW

REVIEWER	Reviewer name: Dr. Peter Flom Institution and Country: Peter Flom Consulting, United States Competing interests: None
REVIEW RETURNED	29-May-2021

GENERAL COMMENTS	The authors have addressed my concerns and I now recommend publication

VERSION 2 – AUTHOR RESPONSE